Progressivity and redistribution of wealth taxation in Spain

A microsimulation analysis using tax register data

Olga Cantó[1,2] · **Francisco García-Rodríguez**[1,2]

[1]Universidad de Alcalá · [2]EQUALITAS

IARIW General Conference · Brussels · 28 August 2026 · 4:00–5:30pm · Room A2

Session: Wealth and fiscal and monetary policy

Presented by the discussant: **Max Ghenis** (PolicyEngine)

Author response: Olga Cantó & Francisco García-Rodríguez

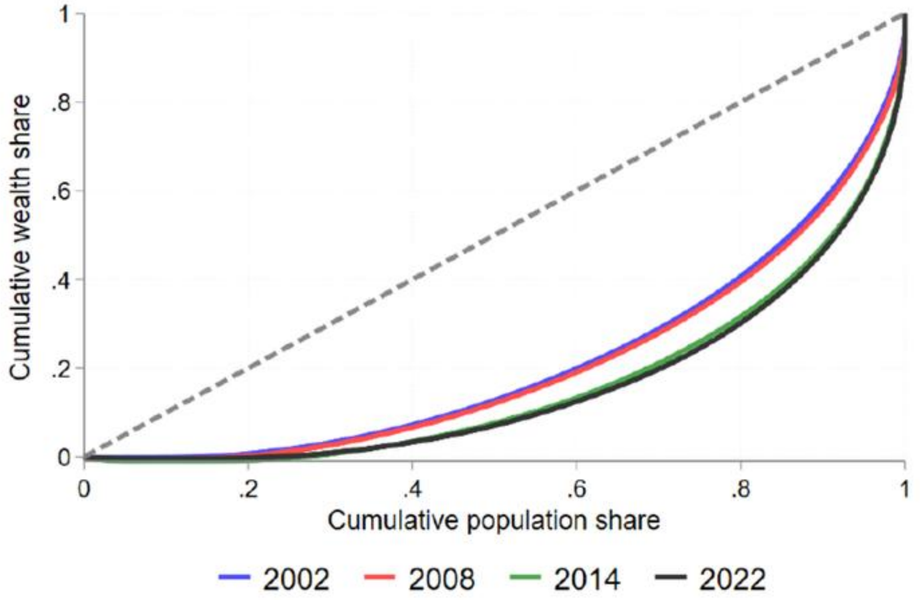
Wealth concentration in Spain has been rising

And most countries have walked away from the tax that targets it

Survey of Household Finances (EFF): the **top 1% share** of net wealth rose from **13.3% (2002) to ~21% (2022)**; the bottom 50% fell from 12.8% to 7.1%.

Meanwhile the annual net wealth tax has nearly vanished: **12 OECD countries in 1990 → 3 by 2023** (Spain, Norway, Switzerland). France retreated to real estate only (ISF → IFI, 2018).

Spain is the natural laboratory: progressive rates, broad legal base, and **17 regional regimes** that can — and do — hollow the tax out.



Lorenz curves of equivalized net wealth (EFF, 2002–2022) — their Figure 1

The Spanish wealth tax: progressive on paper, hollowed out in practice

Impuesto sobre el Patrimonio (IP) — state schedule, regional control

Statutory design (2022 baseline)

- Net wealth above a **€700,000 general exemption** + €300,000 for the main residence
- Progressive state schedule, **0.2% → 3.5%**
- Joint IRPF–IP cap: combined liability ≤ 60% of the income-tax base, with a 20% minimum wealth-tax payment

A tax with a strange life

Introduced 1977 as "temporary"; permanent 1991; suspended 2008; reinstated 2011 — formally temporary, renewed every year since.

Regional autonomy is the story

- Since 2011 regions set rates, thresholds, and rebates — **Madrid and Andalusia: 100% rebate in 2022**
- Catalonia, Extremadura, Valencia keep high effective burdens → horizontal inequity and fiscal competition

The result, in their 2022 data:

170,404 effective taxpayers

of 44.9M individuals (0.4%) · €1.13bn revenue · zero measurable Gini reduction

Scope note: the 2022 state "solidarity" surcharge on large fortunes (ITSGF) is deliberately excluded from the baseline (their footnote 2) — it returns in the discussion.

This paper: what would a different design deliver?

Hold the 2022 population and wealth fixed; swap the tax law; compare

Questions

- How would alternative designs change progressivity and revenue?
- How much redistribution could the revenue buy?
- Which designs bend long-run wealth concentration?
- Who would actually pay, under each?

Contributions

- First wealth-tax design comparison on Spanish **administrative register data**
- Decomposition of redistributive change: **level (incidence) vs distance (progressivity)**
- Long-run (25–100 year) cumulative framework
- Taxpayer profiles under each design (logit)

The comparison set is unusually rich: three real foreign systems (Norway, Switzerland, France), three internal reforms, and Piketty's (2014) normative schedule — all run on the same microdata.

Data: the tax agency's Household Panel

Administrative records for filers and non-filers, compiled by AEAT and provided through the Instituto de Estudios Fiscales

Panel de Hogares (PH), 2022 wave

- **~950,000 households · 2.7M individuals** sampled; weighted to 17.7M households / 44.9M individuals
- Links income-tax returns, wealth-tax declarations, and third-party reports (employers, banks) — covers **non-filers** too
- Panel structure from 2016; used here as a 2022 cross-section

The valuation fix

Real estate is recorded at cadastral value → rescaled to market value with an EFF-based ratio of **×3.08** (uniform across properties).

2022 headline statistics (their Table A.1)

Mean / median net wealth, all individuals	€160,540 / €60,372
Wealth-tax filers (weighted)	170,408
Mean / median net wealth among filers	€2.05M / €1.43M

Limits the authors flag

- Business assets recorded only for wealth-tax filers
- Liabilities not decomposed (EFF used as complement)
- Cadastral values lag markets, unevenly across areas

Baseline IP: Spain's steep state schedule

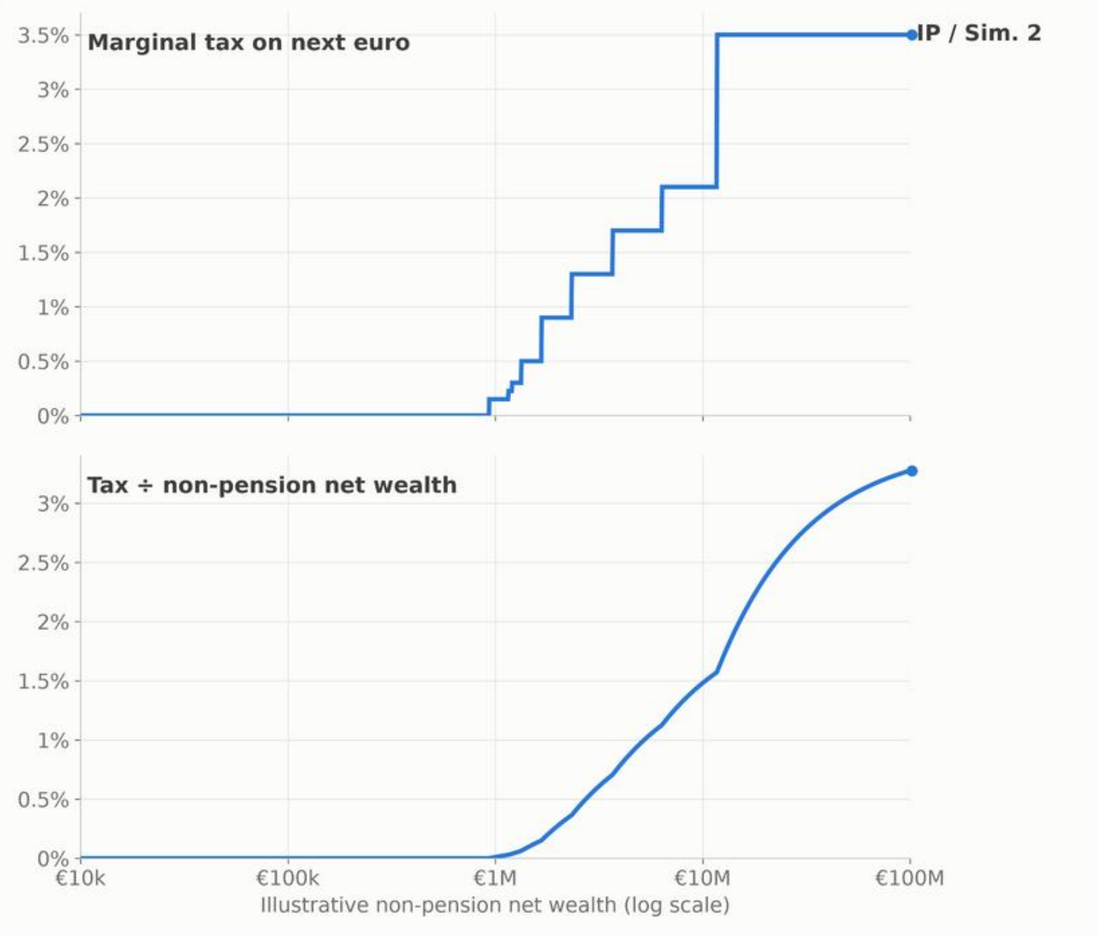
Design 1 in paper order — common illustrative portfolio, single filer

Base and unit
Non-pension net wealth: 25% main residence, 75% other assets. Apply the €300k residence relief plus €700k general exemption; regional rules and the income cap remain outside this chart.

Rates
0.2% → 3.5% over eight brackets — the steepest top rate of the designs you are about to see.

Link to the paper
The baseline actually models 17 regional regimes. This state line is also the later reference for Sim. 2; the paper does not publish its full regional parameter map.

Shape
Late but fierce: the state schedule reaches the highest top rate. The paper explains why far less survives once regions, rebates, and the cap apply.



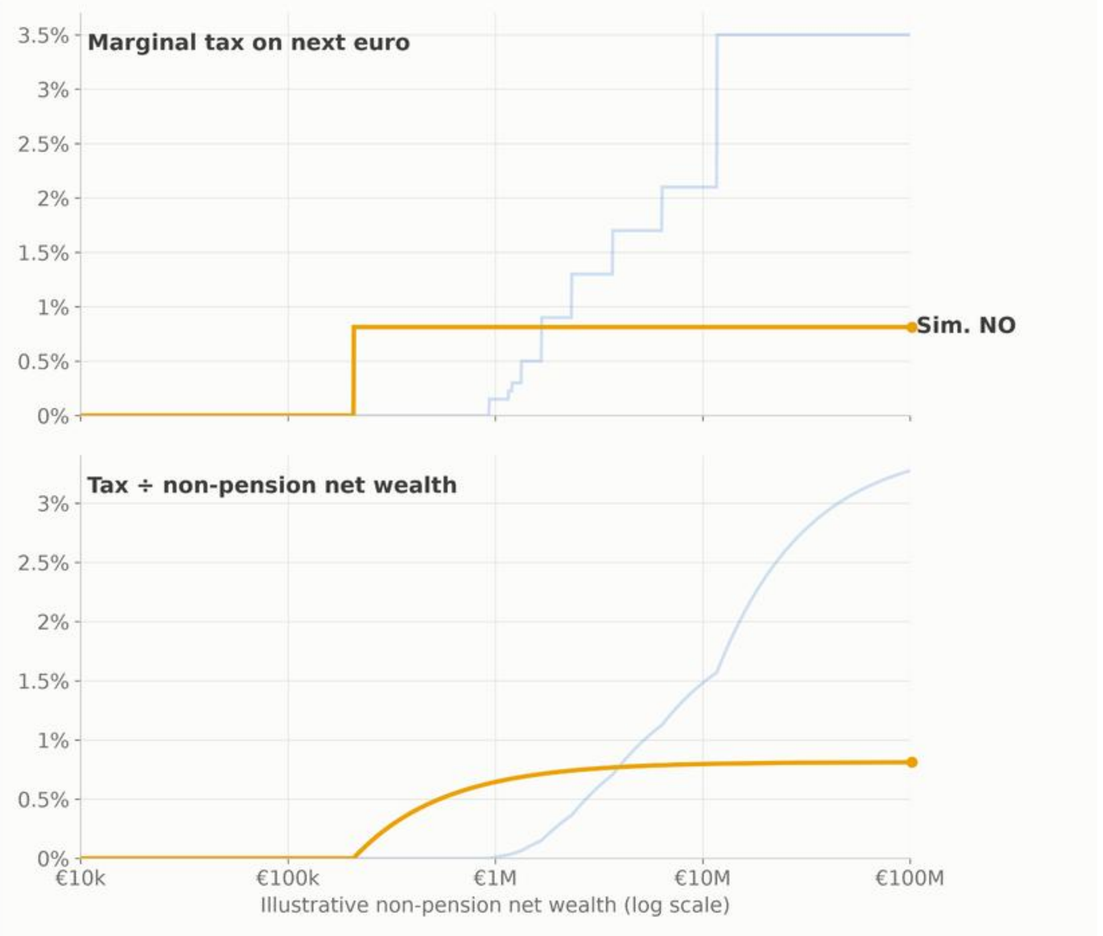
Stylized Norway: a flat 1% after a low threshold

Design 2 in paper order — the paper's Sim. NO, not Norway's two-rate 2022 statute

Base and unit
About €160k of assessable wealth exempt for an individual; double for married couples. Residence is 25% of our portfolio and the paper values that residence at 25%; pensions are outside the path.

Rates
A true flat 1% on the modeled base. Along this portfolio, 81.25¢ of each extra wealth euro is assessable, so the charted marginal burden is 0.8125% after entry.

Shape
Broad and proportional. Actual 2022 law used 0.95% / 1.10%; the paper deliberately stylizes it as flat 1% with housing kept at 25%.



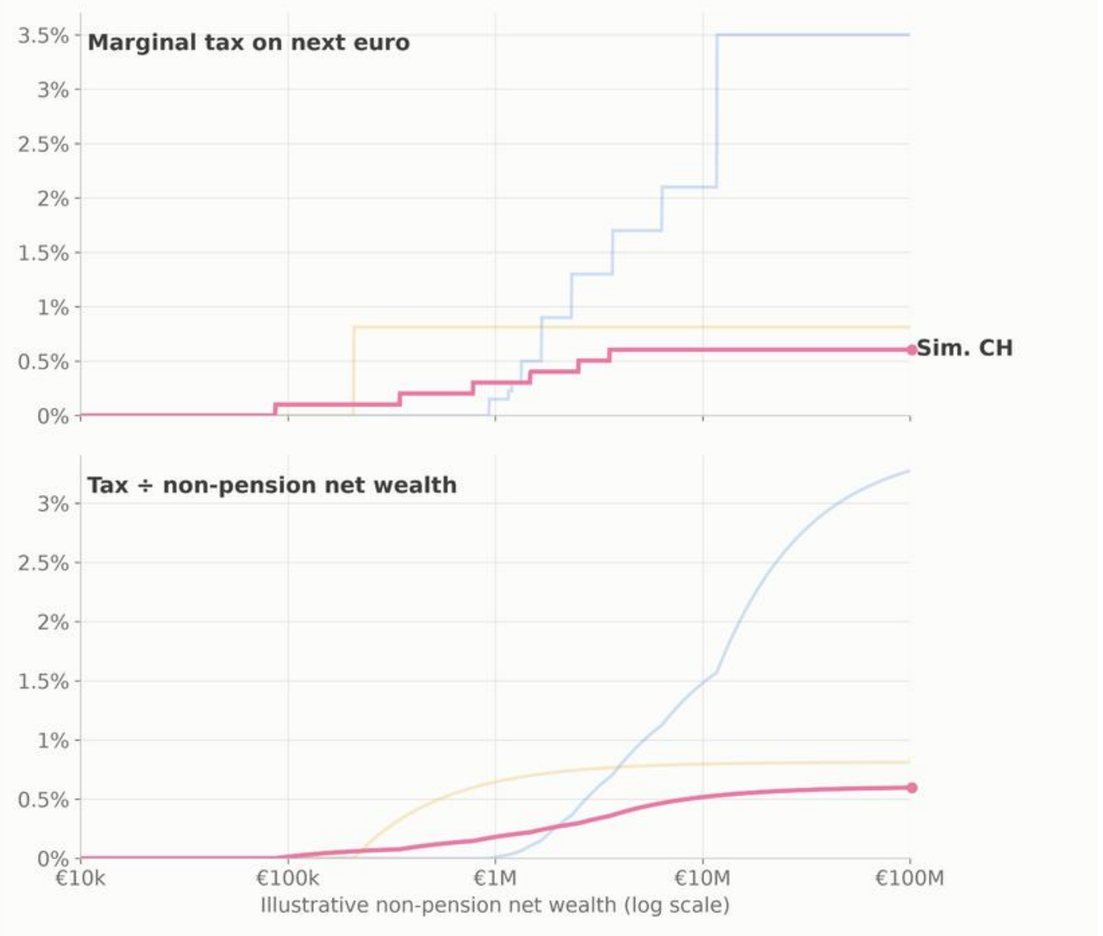
Canton of Zurich: the broad, gentle reference

Design 3 in paper order — illustrative Zurich-city benchmark; paper omits vintage and multiplier

Base and unit
Zurich values the 25% residence share at 70% and other assets in full: 92.5% of this non-pension wealth path before its allowance. Shown allowance: CHF 81k; ECB 2022 average €1 = CHF 1.0047.

Rates
2026-indexed tariff 0–3‰ ×2.18 using canton (99%) + City of Zurich (119%) factors documented for 2022–23 → top ≈0.65%. This is a mixed-vintage illustration.

Shape
Bites almost immediately, climbs slowly, never gets steep — a transparent Zurich-city reference, not a recovered paper parameter. Spouses are jointly taxed.



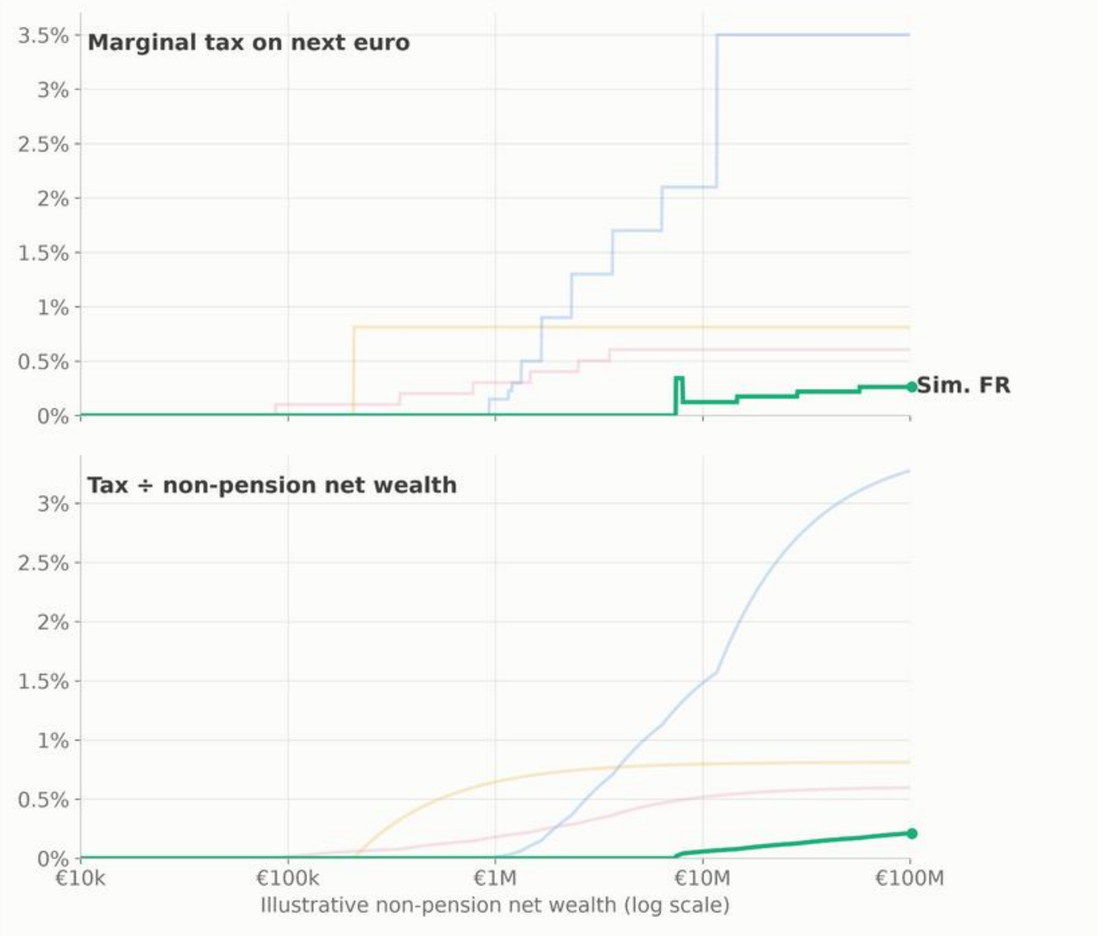
France: a narrow base delays the tax

Design 4 in paper order — the paper's real-estate-only Sim. FR

Base and unit
Only the 25% residence share enters IFI; its 30% main-home abatement leaves 17.5% of total non-pension wealth assessable. Other assets are excluded.

Rates
IFI triggers when assessable property exceeds €1.3M — about €7.4M total wealth on this path. The 0.5–1.5% schedule then counts from €800k.

Shape
Nothing for most of the range, then an ≈€1,250 entry liability, then modest progression — the paper's narrowest base.



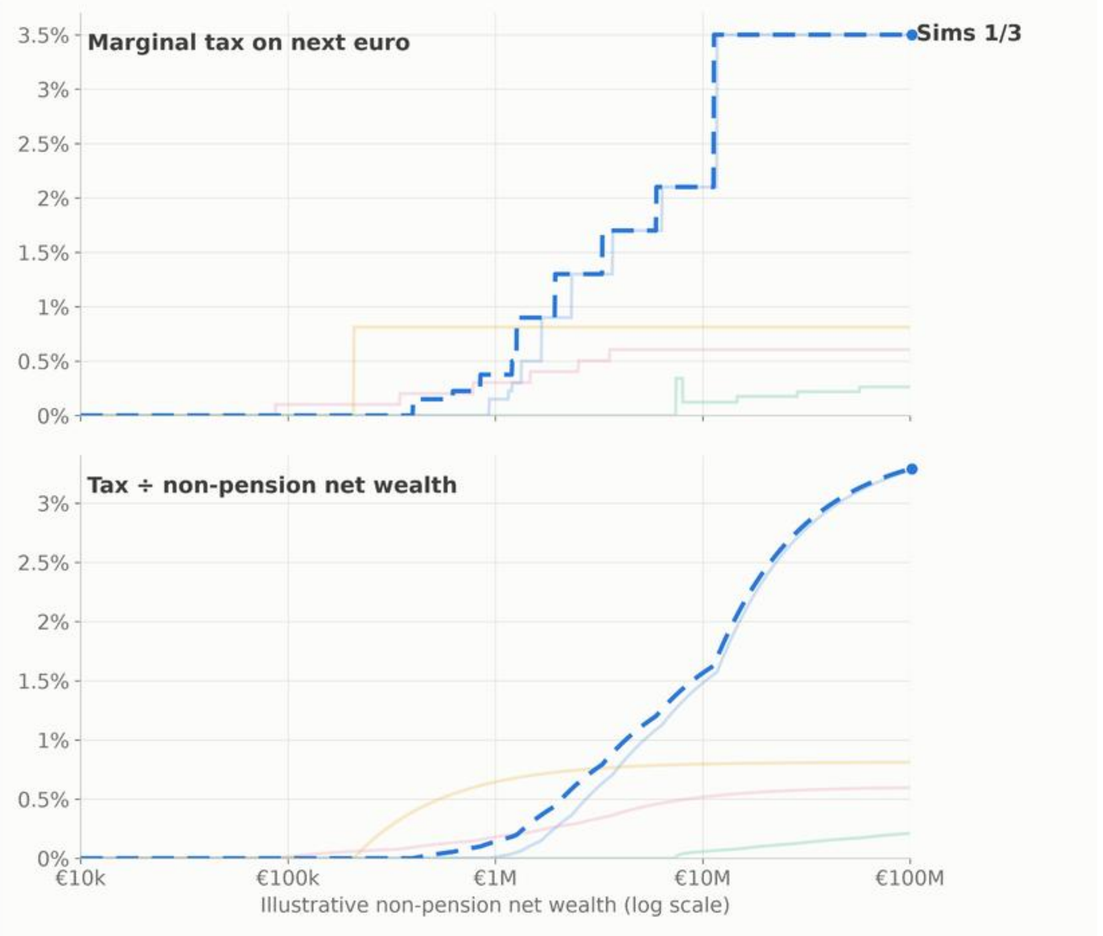
Three Spanish reforms reuse two schedule paths

Designs 5–7 in paper order — Sim. 1, Sim. 2, then Sim. 3

Map the rows
Sim. 1 cuts the general exemption to €300k. Sim. 2 keeps €700k but removes regional deductions and rebates. Sim. 3 combines both.

Two visible paths
The dashed €300k line references Sims 1 and 3; the earlier solid €700k line references Sim. 2. These plotted references use the state 0.2%–3.5% brackets and €300k residence relief; the paper does not publish the exact regional schedule map.

What the paper finds
Effective taxpayers: 1.8M under Sim. 1, 201,687 under Sim. 2, and 4.6M under Sim. 3.



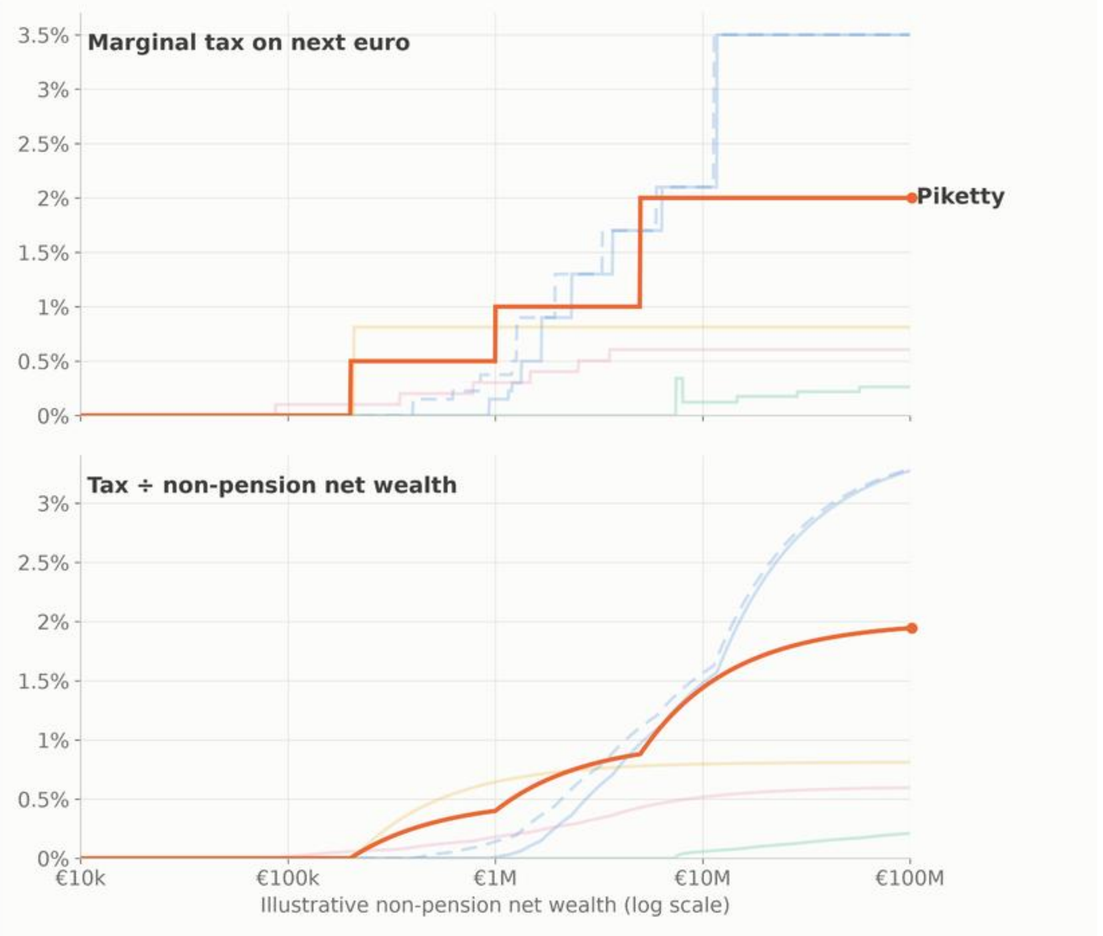
Piketty: the normative benchmark

Design 8 in paper order — the paper's normative benchmark, not enacted law

Base and unit
The paper calls this a full net-wealth base beginning at €200k, but does not publish its asset-level or pension treatment or whether an income cap carries over. The chart assumes full non-pension wealth.

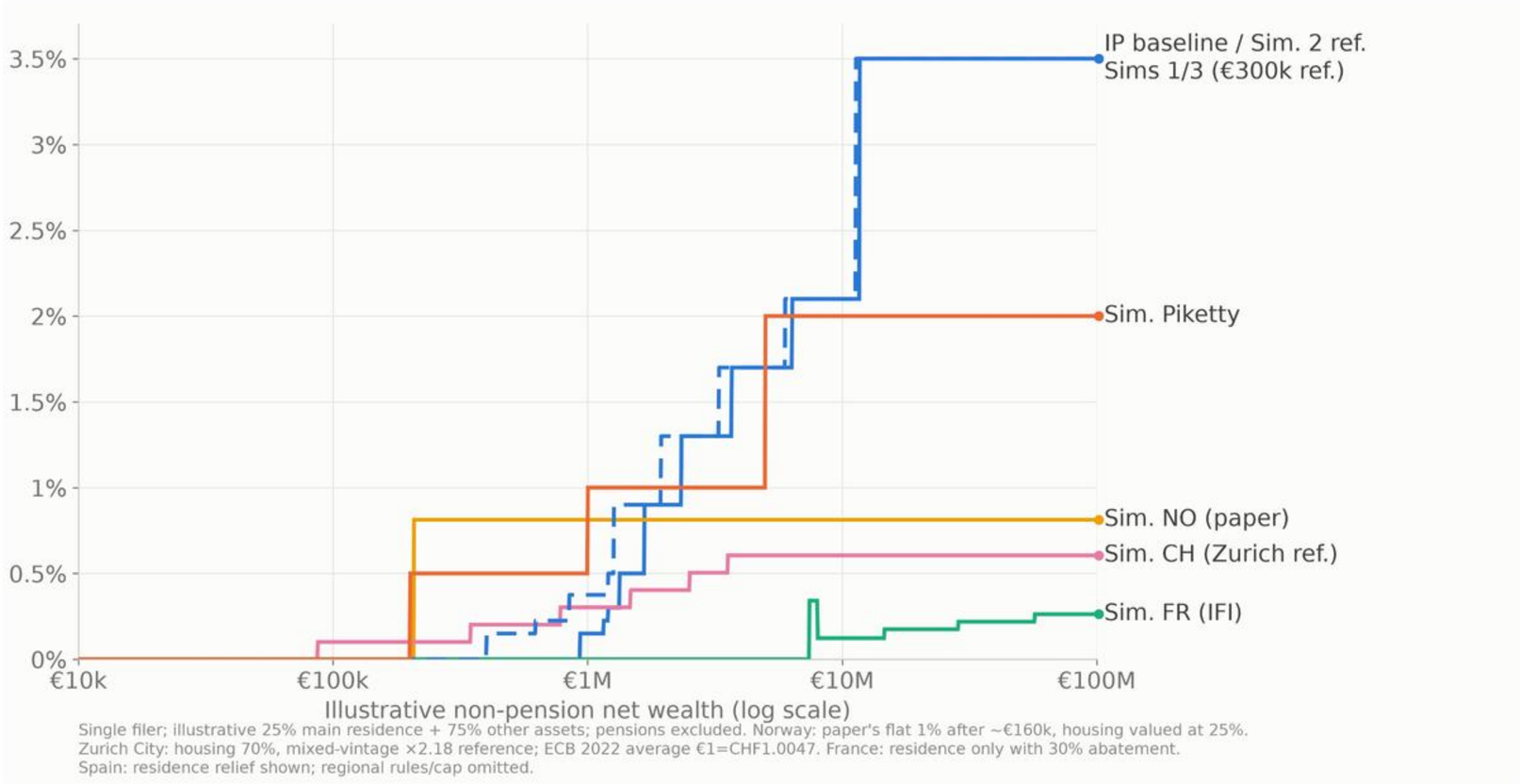
Rates
0.5% / 1% / 2% above €200k / €1M / €5M (the book also floats 0.1% below €200k and 5–10% at the very top).

Shape
Smoothly progressive from modest wealth up. Note how it shadows the dashed €300k Spanish line through the middle of the distribution.



One portfolio makes valuation choices visible

Paper order; 25% main residence + 75% other assets; pension rights excluded



Norway's flat 1% becomes 0.8125% per next wealth euro after its 25% housing valuation; France does not enter until about €7.4M on this property-light path. Spain still reaches the steepest top rate. The table next maps these six paths back to all eight simulations.

The paper turns six schedule paths into eight designs

Paper order: baseline → foreign designs → Spanish reforms → Piketty; same 2022 microdata, no behavioral response

Design	What it is	Threshold / relief	Rates
IP (baseline)	Spanish wealth tax as legislated for 2022, all 17 regional regimes	€700k general; ≤€300k home	0.2–3.5% progressive
Sim. NO	Norwegian design: broad base, residence at 25% of market value	~€160k assessable; 2× married	flat 1% (0.3% + 0.7%)
Sim. CH	Swiss design (Canton of Zurich): real estate at 70% of market value	cantonal allowance	progressive, low rates
Sim. FR	French IFI: real-estate wealth only	€1.3M property trigger	0.5–1.5% progressive
Sim. 1	Spanish IP with the general exemption cut to €300k	€300k general; ≤€300k home	IP schedule
Sim. 2	Spanish IP without regional deductions and rebates	€700k general; ≤€300k home	IP schedule
Sim. 3	Sim. 1 + Sim. 2 combined	€300k general; ≤€300k home	IP schedule
Sim. Piketty	Paper's Piketty-inspired variant; base details incomplete	€200k threshold	0.5% / 1% / 2%

Tables 1 (p.11) and A.2 (p.40). The paper describes pensions as exempt under NO/CH but does not publish the asset-level implementation; its Piketty base is likewise incomplete. Foreign designs drop Spain's cap; IFI has its own 75% income plafonnement.

How they measure it

Standard fiscal-incidence toolkit, plus a decomposition and a long-run extension

Progressivity & redistribution

- **Kakwani** $K = C_t - G_{pre}$: is the burden more concentrated than wealth?
- **Suits** index (inequality-level-free cousin)
- **Reynolds–Smolensky** $R–S = G_{pre} - G_{post} = t/(1–t)\cdot K - D$
- Measured on net wealth *and* equivalized market / disposable income

Level vs distance

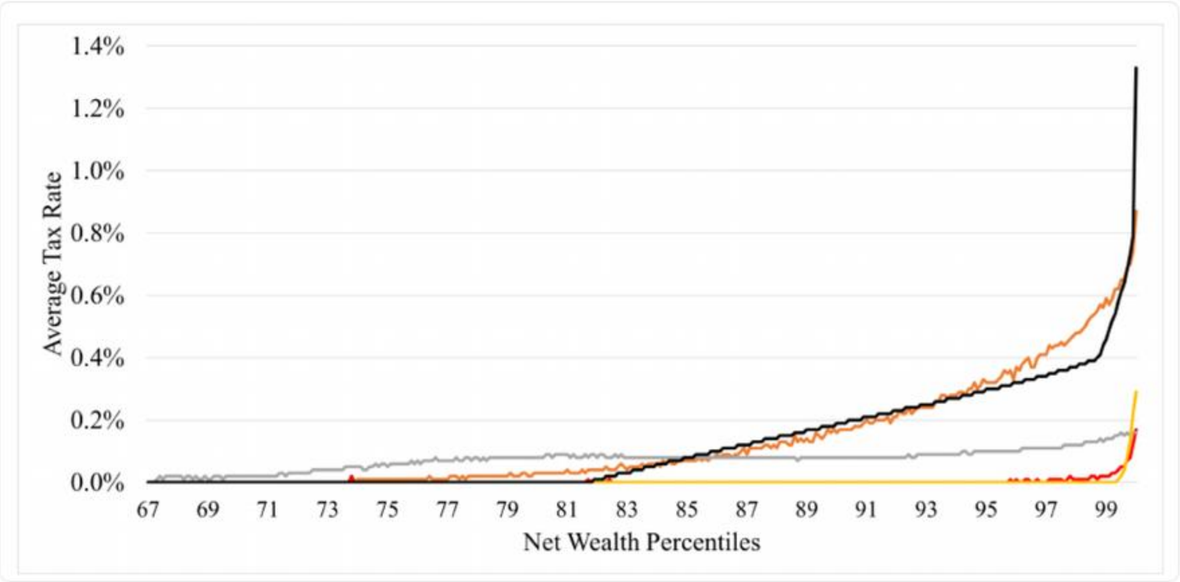
Changes in R–S across designs split into a **level effect** (more incidence — the tax simply collects more) and a **distance effect** (more structural progressivity), following Díaz et al. (2010, 2011).

Long run

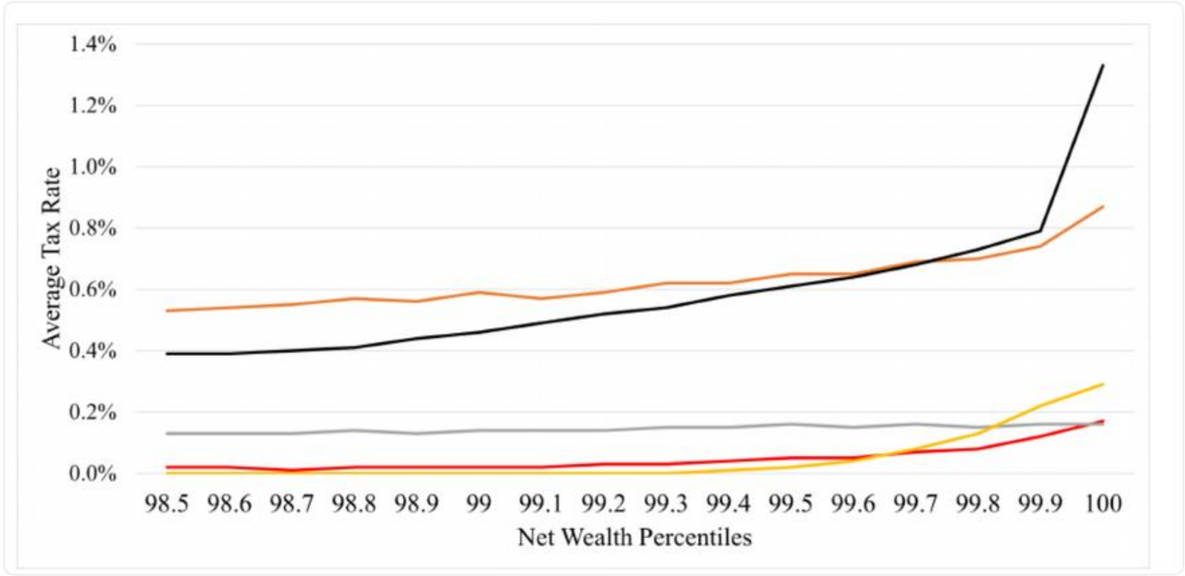
Wealth is a stock: project each person's wealth forward at a constant growth rate $g \in \{1\%, 2\%, 3\%\}$ with and without the annual tax, for T = 25–100 years (Durán-Cabré & Esteller-Moré framework). Compounding turns tiny annual rates into large cumulative wedges.

Result 1 · Who faces the tax, and at what rate

Level measure · no-IP line = 0 Average effective tax rates by net-wealth percentile, 2022 — their Figure 3



From percentile 67



From percentile 98.5

IP Sim. NO Sim. CH Sim. FR Sim. 1
Sim. 2 Sim. 3 Sim. Piketty

Current IP (red): flat zero until ~p99.3. Norway starts near p80; Piketty and Sim. NO climb steepest at the top.

Result 2 · Base breadth, not rate steepness, drives everything

Gini comparator · no IP Individual net wealth, 2022 — condensed from their Table 3 (p.21)

Design	Taxpayers	Revenue (€M)	Avg. rate t	Kakwani K	Suits S	Wealth Gini cut
IP (baseline)	170,404	1,126	0.02%	0.231	0.806	0.000%
Sim. NO	7,647,459	15,456	0.29%	0.191	0.522	0.066%
Sim. CH	12,400,000	4,843	0.09%	0.120	0.276	0.013%
Sim. FR	117,585	1,444	0.03%	0.244	0.893	0.013%
Sim. 1	1,834,488	3,709	0.07%	0.226	0.752	0.013%
Sim. 2	201,687	2,405	0.04%	0.234	0.849	0.013%
Sim. 3	4,626,929	12,673	0.24%	0.231	0.779	0.066%
Sim. Piketty	11,010,000	15,861	0.30%	0.197	0.570	0.080%

Revenue spans 14× — €1.1bn (IP) to €15.9bn (Piketty).

Formal progressivity falls as bases broaden: K and S are highest for IP/FR/Sim. 2.

Short-run Gini moves are tiny for wealth — the action is in revenue and the long run.

Result 3 · Incidence dominates the redistribution gains

Comparator · current IP Change in Reynolds–Smolensky, split using Díaz et al. (2010, 2011) — their Figure 4

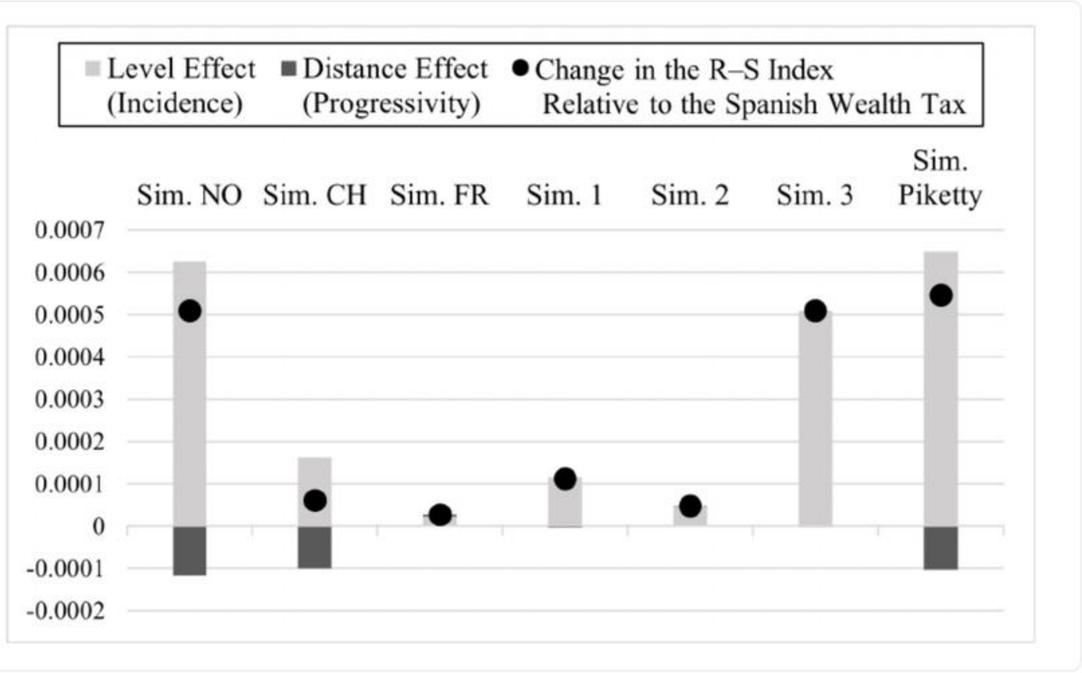
Every alternative redistributes more than the current IP — and in every case the gain is **dominated by the level effect**: the tax simply collects more.

The distance contribution is **negative** for Norway, Switzerland, and Piketty—even though all three taxes remain progressive in absolute terms.

Mechanically: the mean net-of-tax change goes to “level”; the change in normalized pairwise wealth gaps goes to “distance”. This is an accounting split, not a behavioral one.

> Negative is relative, not regressive: Sim. CH has K = 0.120, versus 0.231 for the current IP.

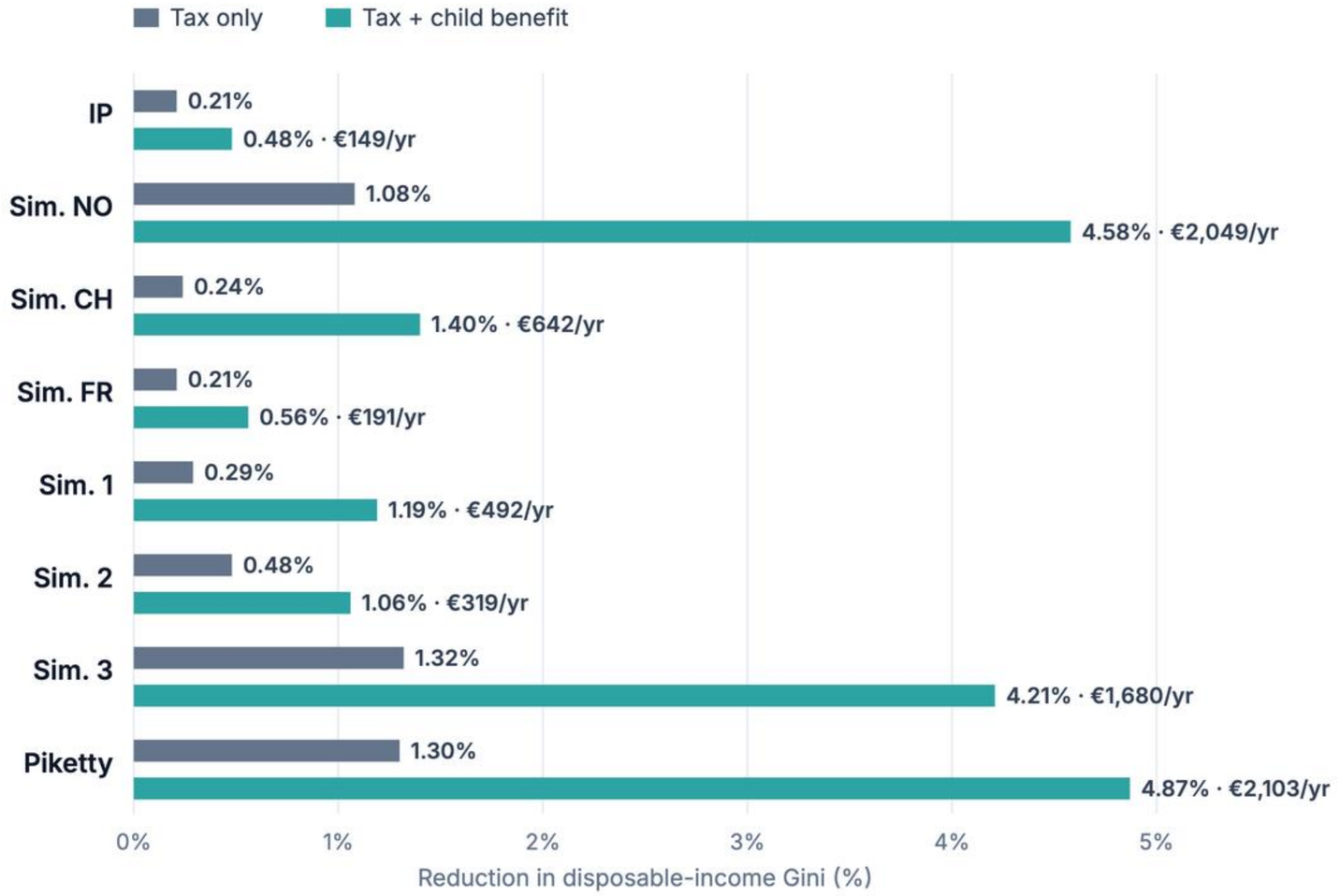
> Their own words: improvements are “driven primarily by the Level Effect” (p.22).



ΔR–S vs the Spanish IP, individual net wealth

Result 4 · Most of the immediate income effect comes from recycling

Comparator · no IP / no transfer Tax only (Table 3, p.21) versus tax + equal annual per-child transfer (Table 4, p.25)



Comparing their tables, tax + benefit cuts the Gini **2.2–5.9×** as much as the tax alone.

For Norway, Sim. 3, and Piketty, the child benefit supplies **69–76% of the combined one-year reduction**.

Derived from rounded R–S values; the paper reports no transfer-only counterfactual.

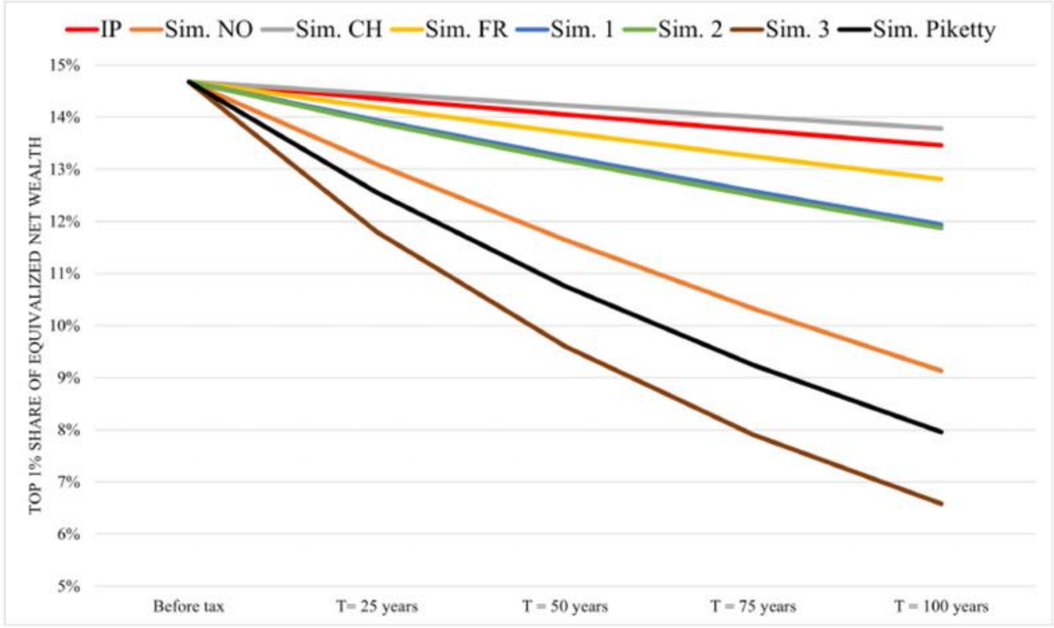
Result 5 · Compounding turns small rates into large long-run wedges

Reference · common T = 0 wealth Top 1% share under repeated annual taxation, g = 2% — their Figure 5

Current IP: top 1% share drifts from 14.7% to **13.5% after 100 years** (−1.2pp).

Norway-style and Piketty schedules: **below 8–9% by T=100** (−5.5 to −6.7pp); Sim. 3 falls furthest among Spanish options (to 6.6%, −8.1pp).

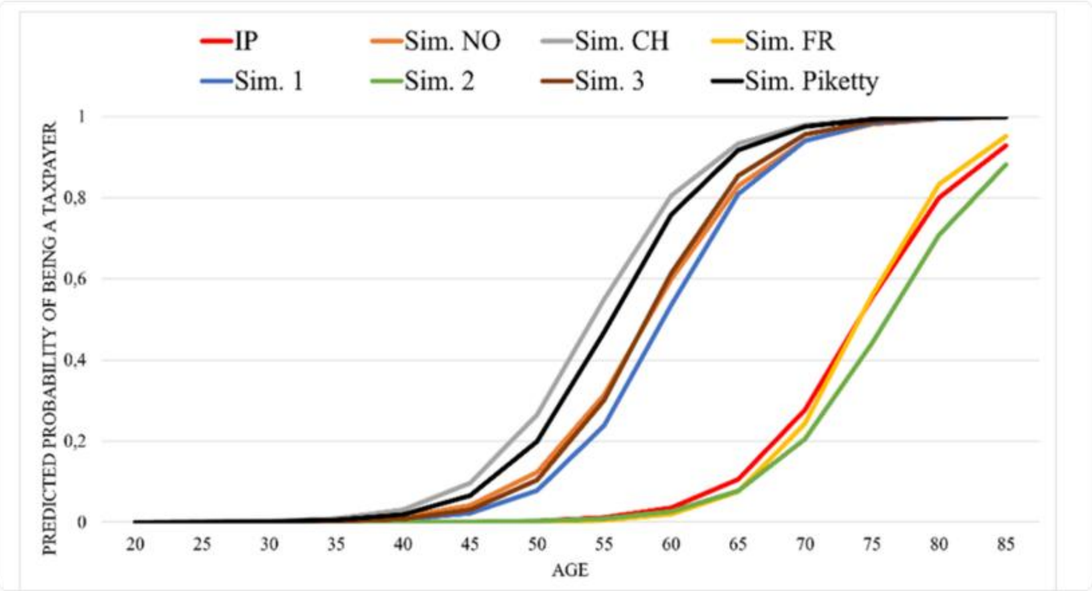
Mechanical projection, not a forecast — the authors present it as illustrating how annual stock taxation cumulates (p.25). Effective long-run rates reach ~22% of wealth for the broad designs (Table 5).



Top 1% share of equivalized net wealth, T = 0 → 100 years

Result 6 · Design choices reshape who pays

Level measure · not Δ vs IP Taxpayer profiles across designs — their Figures 6–7 and Table 6



Predicted probability of paying, by age (logit with region and household controls)

Under the current IP and the French design, payment is an **old-age phenomenon** — probabilities take off only after ~65.

Broad bases (Norway, Switzerland, Sim. 1/3, Piketty) pull the curve left: **younger and more diverse payers** enter the base.

Women are less likely to pay under every design — the administrative footprint of the gender wealth gap.

Their conclusions

As stated in the paper (p.33–34) and their own deck

Current design (IP)

High formal progressivity, minimal effect: 0.4% of individuals pay, €1.1bn raised, no measurable inequality reduction; regional rebates create horizontal inequity.

Reformed designs (Sim. 3, Piketty, NO)

Broader bases and harmonized rules multiply taxpayers (4.6–11M) and revenue (€12.7–15.9bn), cut disposable-income inequality once recycled, and bend long-run wealth concentration.

> The policy claim: **redistributive gains come from broadening the base and raising effective incidence — not from steepening marginal rates.**

Discussion

Four comments from a tax-microsimulation seat

1 · Use the outturns — separate model fit from policy change

Benchmark against the same tax year; use 2023 as a partial test of Sim. 2

SAME-YEAR AGGREGATE CHECK

€1.126bn → **€1.250bn**

Simulated 2022 IP — Observed 2022 IP

The model is **10% below** AEAT's nationwide total; population-scope differences may explain part of the gap.

AFTER MADRID CHANGED ITS REBATE RULE

	2022		2023	Change
IP	€1.250bn	→	€1.926bn	+€676m
ITSGF	€0.630bn	→	€0.035bn	−€595m
Combined	**€1.880bn**	→	**€1.961bn**	**+€81m**

AEAT attributes **90.6%** of the IP increase to Madrid. Most of the headline jump is a shift between the two taxes, not a baseline forecast miss.

Suggested revision: call the benchmark a **2022 pre-ITSGF counterfactual**, compare it with 2022 IP while noting the population scope, and use 2023 combined IP + ITSGF as a partial check on Sim. 2. Also correct footnote 2: ITSGF first accrued on 31 December 2022.

2 · Add behavioral sensitivity ranges to the static estimates

Static scores are a useful benchmark; a modest bounds exercise would show how robust the ranking is

An illustrative sensitivity range (not a point estimate)

- Advani & Tarrant: base −7–17% per 1pp → **Sim. NO €15.5bn → €12.8–14.4bn**
- Mas-Montserrat: −3.21% per 0.1pp ATR → ~8–10% lower
- Brülhart: ≥43% reported-wealth response per 1pp (high-response anchor)
- Saez–Zucman: 15% / 50% erosion scenarios

Recent evidence reinforces the harmonization case

Agrawal, Foremny & Martínez-Toledano: mobility induced by Madrid's zero rate costs 6× more in income tax than it saves in wealth tax; **Madrid itself would gain from a harmonized positive rate** — evidence consistent with your Sim. 2/3 case.

Two external anchors for context

Norway raises ~0.5% of GDP vs Sim. NO's 1.13% of Spain's. Sims 1–3 also retain the 60% IRPF–IP cap.

Because behavioral responses likely rise with rates, these ranges may **strengthen your breadth-over-rates conclusion** while lowering the headline euros. Showing both figures would make the conclusion more robust.

3 · The long-run mechanics merit a consistency check

Reapply each schedule under an actual or explicitly assumed threshold path, then test whether the long-run ranking survives

Reapply the schedule as wealth grows

At g = 2%, no-tax wealth grows **7.24×** by T=100. But each year-0 average rate stays fixed: nobody enters, exits, or changes bands. A schedule-based rerun could materially change the paths.

The equivalized long-run Gini

Uniform growth leaves the no-tax Gini unchanged. The printed inputs imply at most **~0.17pt down** by T=25; Table 5 reports **+1.04pt**. Could definitions or equivalization explain it?

Reconcile total net wealth: €5.3T vs €7.2T

The reported t and $t_{s/r}$ imply **≈ €5.3T**, while Table A.1's mean × count implies **€7.2T**. Which aggregate seeds the projection?

Threshold paths and comparators differ by design

Spain: nominal until amended · Norway: annual Storting decision · Zurich: bracket amounts CPI-adjusted every two years. State each path—and label Tables 3–4 vs no IP, Figure 4 vs current IP.

4 · Lead with the strongest finding — and make it reproducible

The breadth–progressivity trade-off is the paper's most interesting result; replication is one disclosure package away

Framing

The broad-base designs redistribute more even as Kakwani falls: 0.231 → 0.191 (Norway) and 0.197 (Piketty). The level effect dominates. That breadth–incidence trade-off is a sharper finding. The baseline is also 8% low on payers and 10% low on revenue versus AEAT's 2022 totals — a useful two-moment check.

Replicability

Researchers can request the Panel de Hogares free under the IEF protocol. The remaining author-controlled inputs are code, implementation-ready parameters, and the asset-base map. Zurich shows why: its 0.3% statutory top tariff becomes ~0.65% only after canton and city multipliers.

A practical package: publish the 17-region + 3-foreign-design + Piketty parameters, the asset-by-asset base map, and calculator code. **We've started an open re-encoding of all eight designs** — with those disclosures and a PH extract, it could become a full replication.

Questions for discussion

Could you clarify the base? Which property values, pension rights, business assets, and liabilities enter each scenario? What does Piketty raise on one common asset base?

Could 2023 be an out-of-sample test? Madrid un-rebated + ITSGF is a partial dress rehearsal of Sim. 2 — does the combined ~€1.96bn outturn match what your model predicts?

How is the cap applied? Is the 60% IRPF–IP limit active in Sims 1–3 and Piketty, and how much revenue does it remove from each?

Thank you — and over to **Olga Cantó** and **Francisco García-Rodríguez**

Max Ghenis · max@policyengine.org · review + open re-encoding: github.com/MaxGhenis/iariw-2026-canto-garcia-replication